ARJUNA JEE 2027

Lecture -08 | Mathematics

SEQUENCE & SERIES

By- Sachin Jakhar Sir



Topics *to be covered*

A **Properties of Sigma**

B **Miscellaneous Series**

C **Question Practice**

D



Question Practice

Question [JEE Mains-2023]

If the sum of the series

$$\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{2^2}-\frac{1}{2\cdot 3}+\frac{1}{3^2}\right)+\left(\frac{1}{2^3}-\frac{1}{2^2\cdot 3}+\frac{1}{2\cdot 3^2}-\frac{1}{3^3}\right)+\left(\frac{1}{2^4}-\frac{1}{2^3\cdot 3}+\frac{1}{2^2\cdot 3^2}-\frac{1}{2\cdot 3^3}+\frac{1}{3^4}\right)+\cdots$$

is $\frac{\alpha}{\beta}$, where α and β are co-prime, then $\alpha+3\beta$ is equal to

$x=\frac{1}{2}$, $y=\frac{1}{3}$

$1+3(2)=7$

7 Ans.

$\frac{\alpha}{\beta}=\frac{1}{2}$

$$(x-y)+(x^2-xy+y^2)+(x^3-x^2y+xy^2-y^3)+\ldots\ldots\infty$$

$$\frac{x^2-y^2}{x+y}+\frac{x^3+y^3}{x+y}+\frac{x^4-y^4}{x+y}+\frac{x^5+y^5}{x+y}+\ldots\ldots\infty$$

$$(x-y)+(x^2-xy+y^2)+(x^3-x^2y+xy^2-y^3)+\ldots\ldots\infty$$

$$\frac{x^2-y^2}{x+y}+\frac{x^3+y^3}{x+y}+\frac{x^4-y^4}{x+y}+\frac{x^5+y^5}{x+y}+\ldots\ldots\infty$$

$$\frac{1}{x+y}\left\{x^2-y^2+x^3+y^3+x^4-y^4+x^5+y^5+\ldots\ldots\infty\right\}$$

$$\frac{1}{x+y}\left\{\left(x^2+x^3+x^4+x^5+\ldots\infty\right)-\left(y^2-y^3+y^4-y^5+\ldots\infty\right)\right\}$$

($x=-y$)

$$\frac{1}{(x+y)}\left\{\left(\frac{x^2}{1-x}\right)-\left(\frac{y^2}{1+y}\right)\right\}=\frac{6}{5}\left(\frac{1}{2}-\frac{1}{12}\right)=\frac{\cancel{6}}{\cancel{5}}\times\frac{\cancel{5}}{\cancel{12}_2}=\boxed{\frac{1}{2}}$$

$$x+y=\frac{1}{2}+\frac{1}{3}=\frac{5}{6}$$

$$\frac{x^2}{1-x}=\frac{\frac{1}{4}}{\frac{1}{2}}=\frac{2}{4}=\frac{1}{2}$$

$$\frac{y^2}{1+y}=\frac{\frac{1}{9}}{\frac{4}{3}}=\frac{3}{9\times4}=\frac{1}{12}$$

Challenger Question [JEE Mains-2021]

Consider an arithmetic series and a geometric series having four initial terms from the set {11, 8, 21, 16, 26, 32, 4}. If the last terms of these series are the maximum possible four digit numbers, then the number of common terms in these two series is equal to

A.P. → 11, 16, 21, 26 .... 256 .... 4096 9991, 9996

50th

G.P. → 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 8192

fnf = 3 ✓

Question [JEE Mains-2015]

If m is the A.M. of two distinct real numbers l and $n(l, n > 1)$ and G_1, G_2 and G_3 are three geometric means between l and n, then $G_1^4 + 2G_2^4 + G_3^4$ equals.

A $4l^2mn$

B $4lm^2n$

C $4lmn^2$

D $4l^2m^2n^2$

$$\frac{l+n}{2} = m \quad \text{—①}$$

l, G_1, G_2, G_3, n → G.P.

$l,\ lr,\ lr^2,\ lr^3,\ lr^4$

$n = lr^4$

$\# \ \frac{n}{l} = r^4$

$$(lr)^4 + 2(lr^2)^4 + (lr^3)^4 = l^4r^4 + 2l^4r^8 + l^4r^{12}$$

$$= l^4r^4(1 + 2r^4 + r^8)$$

$$= l^4r^4(r^4+1)^2$$

$$= l^4 \frac{n}{l}\left(\frac{n}{l}+1\right)^2 = \frac{l^4 n}{l}\frac{(n+l)^2}{l^2} = ln(l+n)^2$$

$$ln(2m)^2 = 4m^2ln$$

Example

The A.M. between two positive numbers exceeds the G.M. by 5, and the G.M. exceeds the H.M. by 4. Find the numbers.

a, b

$AM = \frac{a+b}{2}$

$GM = \sqrt{ab}$

$HM = \frac{2ab}{a+b} = \frac{2}{\left(\frac{1}{a}+\frac{1}{b}\right)}$

$AM = GM + 5$

$GM = HM + 4$

$\frac{a+b}{2} = \sqrt{ab} + 5$

$\sqrt{ab} = \frac{ab}{\left(\frac{a+b}{2}\right)} + 4$

$\sqrt{ab} = \frac{ab}{(\sqrt{ab}+5)} + 4 \Rightarrow \sqrt{ab} = \frac{ab + 4\sqrt{ab} + 20}{\sqrt{ab}+5}$

Ans: 40, 10

fnf.

$$\sqrt{ab} = \frac{ab + 4\sqrt{ab} + 20}{\sqrt{ab} + 5}$$

$$ab + 5\sqrt{ab} = ab + 4\sqrt{ab} + 20$$

$$\sqrt{ab} = 20$$

$$\frac{a+b}{2} = \sqrt{ab} + 5$$

$$a + b = 50$$

$$ab = 400$$

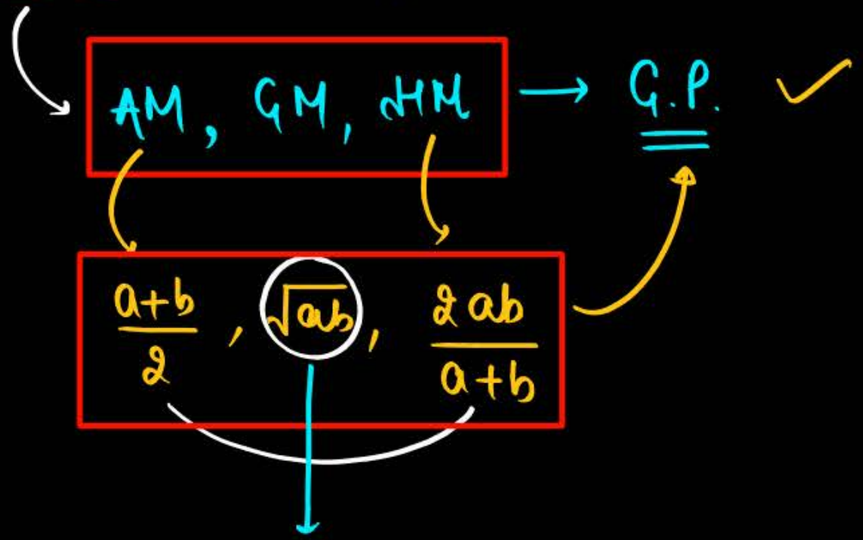
$$40, 10$$

$$b = \frac{400}{a}$$

$$a + \frac{400}{a} = 50$$

$$a^2 - 50a + 400 = 0 \quad (40, 10)$$

For : Two +ve numbers (a, b)



Square $= ab = \frac{a+b}{2} \times \frac{2ab}{a+b}$.

$= ab$.

Example

The A.M. between two positive numbers exceeds the G.M. by 5, and the G.M. exceeds the H.M. by 4. Find the numbers.

a, b

$AM = \frac{a+b}{2}$

$GM = \sqrt{ab}$

$HM = \frac{2ab}{a+b}$

$= \frac{2}{\left(\frac{1}{a}+\frac{1}{b}\right)}$

$AM = GM + 5$

$GM = HM + 4$

$AM = 25$

M-2:-

$GM^2 = AM \cdot HM$

$GM^2 = (GM+5)(GM-4)$

$GM^2 = GM^2 + GM - 20$

$20 = GM$



A.G.P. (ARITHMETIC GEOMETRIC PROGRESSION)

A.G.P. : $a, (a + d)r, (a + 2d)r^2, (a + 3d)r^3, \ldots\ldots \rightarrow$ A.G.P.

Method to find SUM of A.G.P. : Multiply by 'r' $\rightarrow$ Shift $\rightarrow$ Subtract

Example

Find : $S = \frac{3}{5} + \frac{5}{15} + \frac{7}{45} + \frac{9}{135} + \ldots$ **upto infinite terms**

$\frac{1}{3}\times$

$$\frac{S}{3} = 0 + \frac{3}{15} + \frac{5}{45} + \frac{7}{135} + \ldots\ldots\infty$$

(−)

$$\frac{2S}{3} = \frac{3}{5} + \left(\frac{2}{15} + \frac{2}{45} + \frac{2}{135} + \ldots\ldots\infty\right)$$

$$\# \quad \frac{2S}{3} = \frac{3}{5} + \frac{2}{15}\left(1 + \frac{1}{3} + \frac{1}{9} + \ldots\infty\right)$$

Example

Find sum of "n-terms" & S_∞ : $S = \frac{3}{5} + \frac{5}{15} + \frac{7}{45} + \frac{9}{135} + \ldots$

$\times\frac{1}{3}$ $\times\frac{1}{3}$ $\times\frac{1}{3}$

$$S = \frac{3}{5} + \frac{5}{15} + \frac{7}{45} + \frac{9}{135} + \ldots\ldots + T_{n-1} + T_n$$

$\frac{1}{3}\times$

$$\frac{S}{3} = 0 + \frac{3}{15} + \frac{5}{45} + \frac{7}{135} + \ldots\ldots\ldots \quad \frac{T_{n-1}}{3} + \frac{T_n}{3}$$

$\ominus$

$$\frac{2S}{3} = \frac{3}{5} + \frac{2}{15} + \frac{2}{45} + \frac{2}{135} + \ldots\ldots\ldots + \left(T_n - \frac{T_{n-1}}{3}\right) - \frac{T_n}{3}$$

$$S = \frac{3}{5} + \frac{5}{15} + \frac{7}{45} + \frac{9}{135} + \cdots\cdots + T_{n-1} + T_n$$

$\frac{1}{3} \times$

$$\frac{S}{3} = 0 + \frac{3}{15} + \frac{5}{45} + \frac{7}{135} + \cdots\cdots \qquad \frac{T_{n-1}}{3} + \frac{T_n}{3}$$

$\ominus$

$$\frac{2S}{3} = \frac{3}{5} + \frac{2}{15} + \frac{2}{45} + \frac{2}{135} + \cdots\cdots + \left(T_n - \frac{T_{n-1}}{3}\right) - \frac{T_n}{3}$$

$$\frac{2S}{3} = \frac{3}{5} + \frac{2}{15}\left(1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots\cdots\cdots\cdots\right) - \frac{T_n}{3}$$

G.P. $a = 1,\ r = \frac{1}{3}$

n.o.t. $= n-1$

$$\frac{2S}{3} = \frac{3}{5} + \frac{2}{15}\left(1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots\dots\dots\dots\dots\right) - \frac{T_n}{3}$$

G.P. $a = 1,\ r = \frac{1}{3},\ n.o.t. = n$

$$\frac{2S}{3} = \frac{3}{5} + \frac{2}{15}\left(1\left(\frac{1-\left(\frac{1}{3}\right)^{n-1}}{\frac{2}{3}}\right)\right) - \frac{1}{3}\left(T_n\right)$$

Pattern observation

$$\frac{2S}{3} = \frac{3}{5} + \frac{1}{5}\left(1-\left(\frac{1}{3}\right)^{n-1}\right) - \frac{1}{3}\left(\frac{2n+1}{5\times 3^{n-1}}\right)$$

$\times \frac{3}{2}$

$$S = \frac{9}{10} + \frac{3}{10}\left(1-\left(\frac{1}{3}\right)^{n-1}\right) - \frac{1}{2}\left(\frac{2n+1}{5\times 3^{n-1}}\right)$$

→ Sum of n-terms

$$\frac{25}{3} = \frac{3}{5} + \frac{1}{5}\left(1-\left(\frac{1}{3}\right)^{n-1}\right) - \frac{1}{3}\left(\frac{2n+1}{5\times 3^{n-1}}\right)$$

$\times \frac{3}{25}$

$$S_n = \frac{9}{10} + \frac{3}{10}\left(1-\left(\frac{1}{3}\right)^{n-1}\right) - \frac{1}{2}\left(\frac{2n+1}{5\times 3^{n-1}}\right)$$

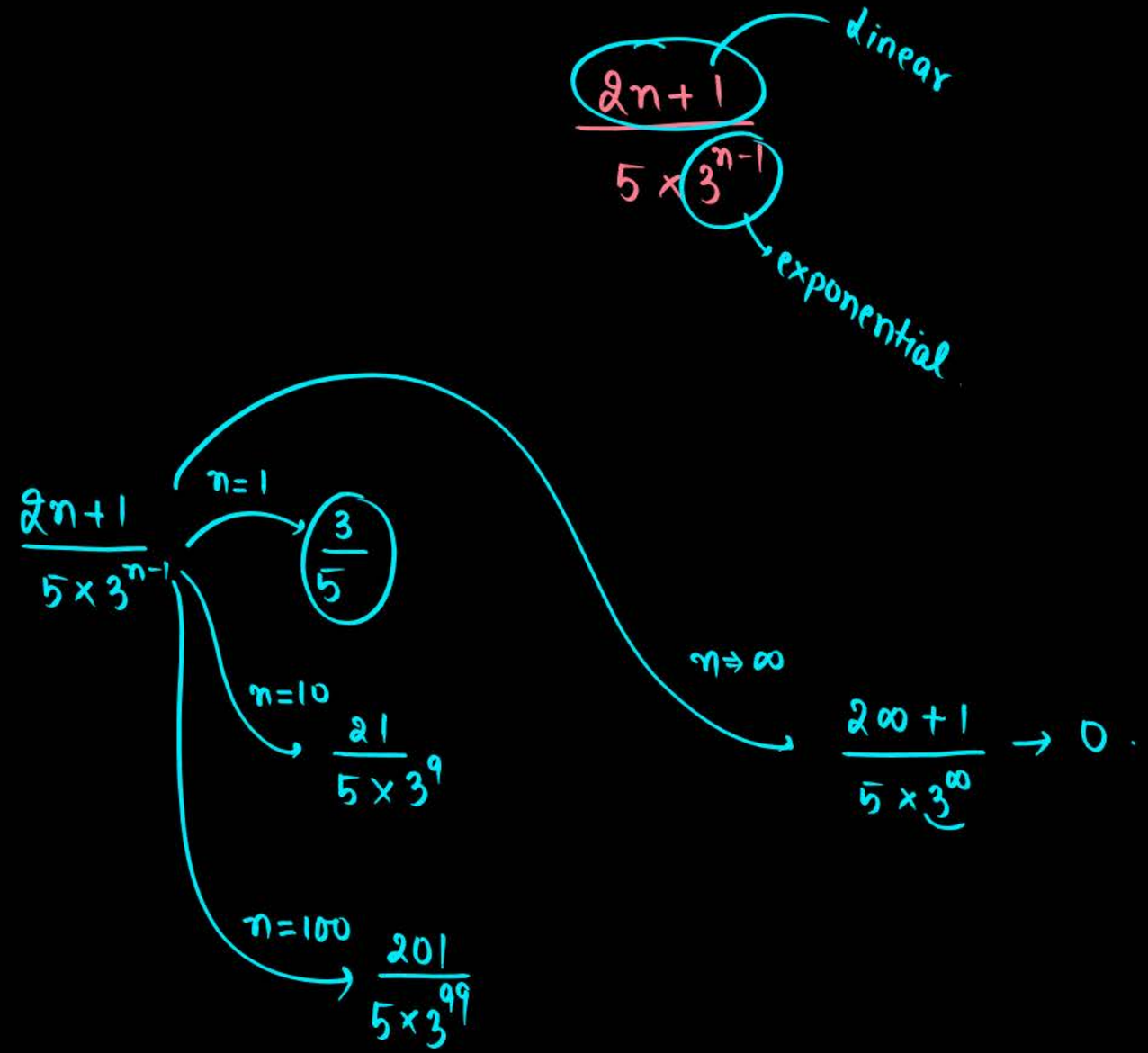
→ Sum of n-Terms

$n \to \infty$

$$S_\infty = \frac{9}{10} + \frac{3}{10}\left(1-0\right) - \frac{1}{2}\left(0\right)$$

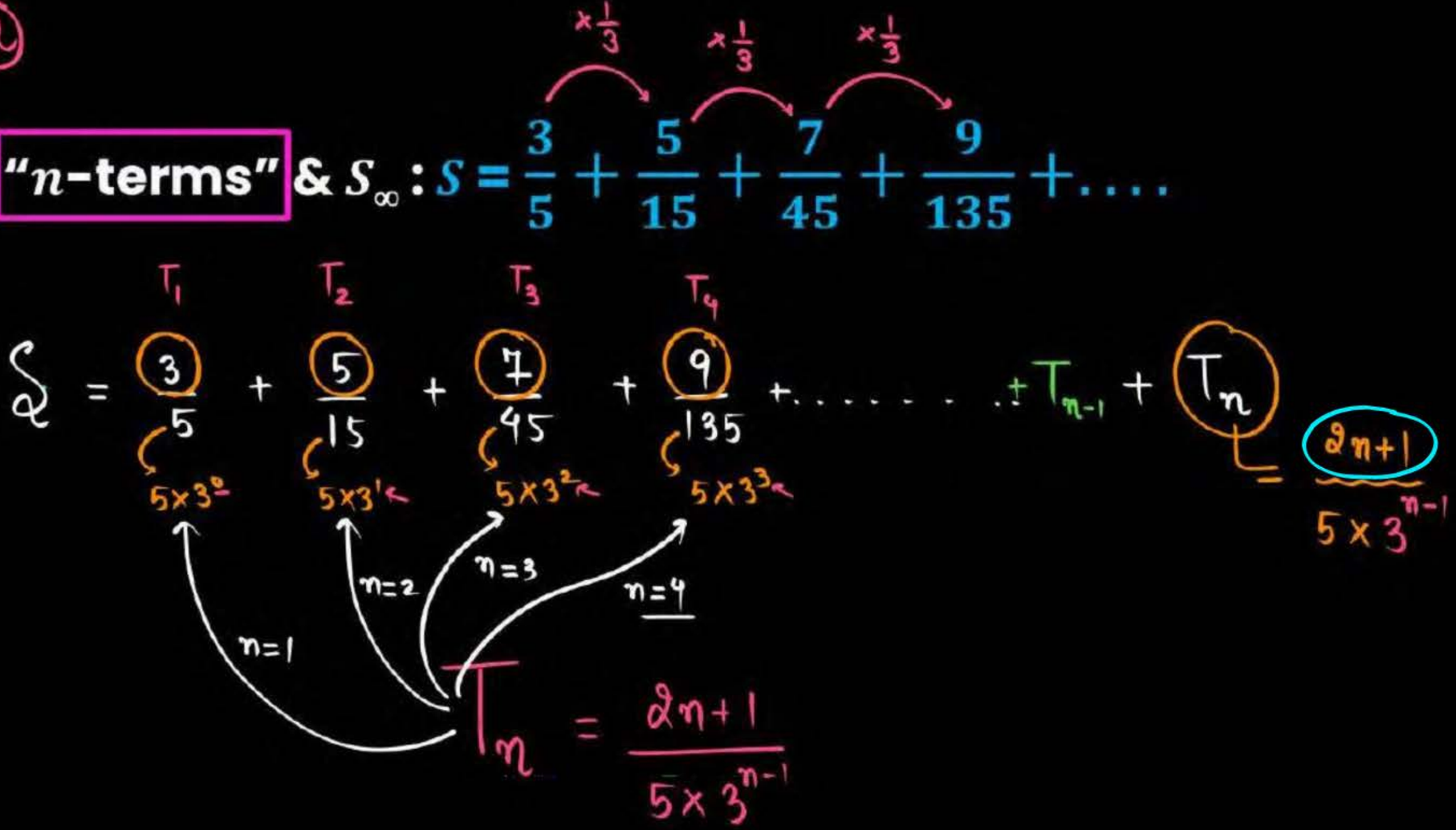
$$\left(\frac{1}{3}\right)^{n-1} = \left(\frac{1}{3}\right)^{\infty} = \frac{1}{3^\infty} = 0$$

$$S_\infty = \frac{12}{10} = \frac{6}{5}$$ ✓


$\frac{2n+1}{5\times 3^{n-1}}$
linear
exponential
$n=1$
$\frac{3}{5}$
$n=10$
$\frac{21}{5\times 3^9}$
$n=100$
$\frac{201}{5\times 3^{99}}$
$n\to\infty$
$\frac{2\infty+1}{5\times 3^{\infty}} \to 0$.

Example (2)

Find sum of "n-terms" & S_∞ : $S = \frac{3}{5} + \frac{5}{15} + \frac{7}{45} + \frac{9}{135} + \ldots$



Blue बादल

Kisi bhi series mein terms ek particular number se multiply ho rhi ho!

⇓

Sum nikalne ka method is as same as A.G.P.

⇓

multiply by r ⇒ shift ⇒ subtract

Question

• Multiple layer A.G.P.

If $|x| < 1$, then compute S_∞ :

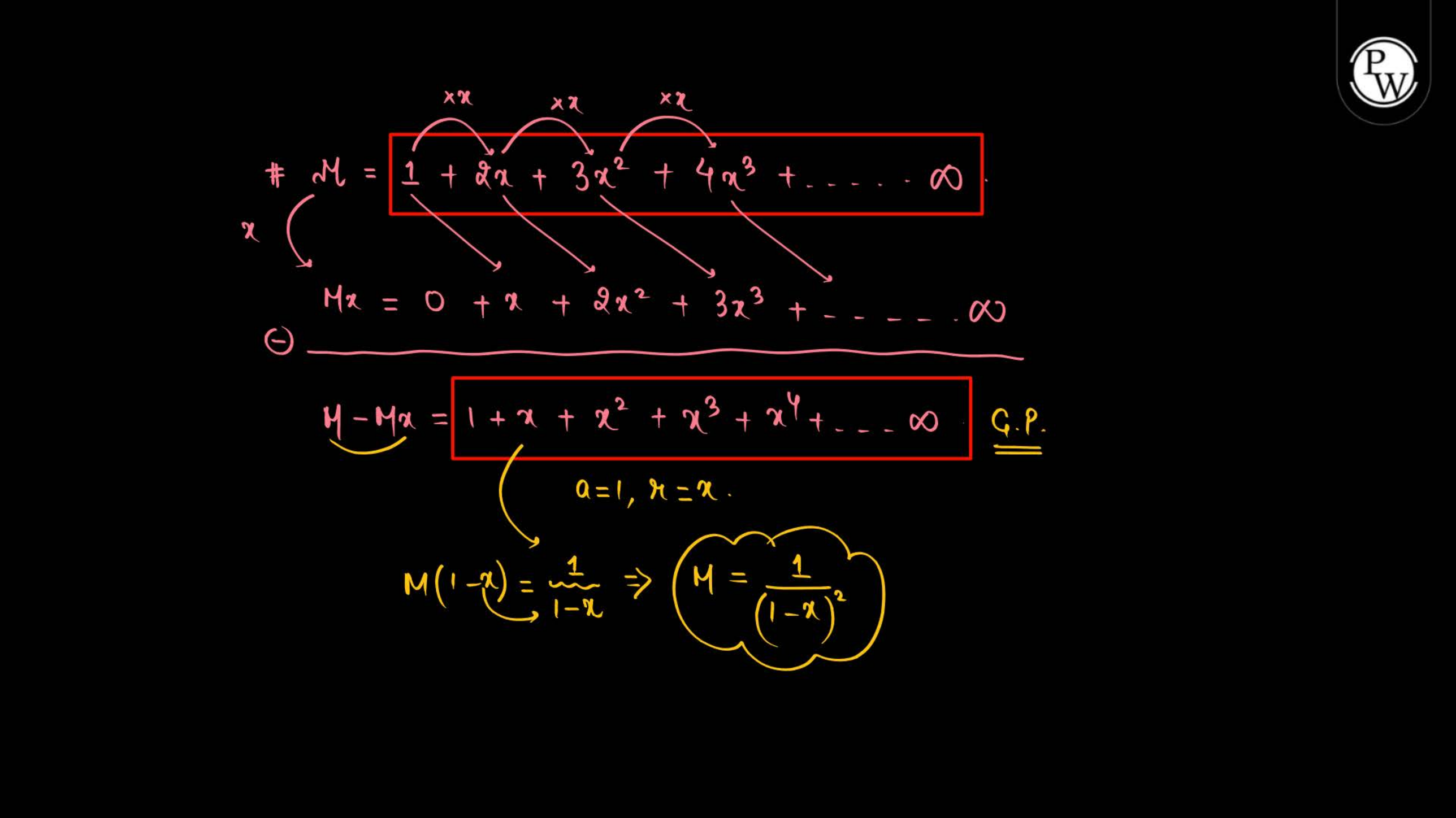
$$S_\infty = 1 + 3x + 6x^2 + 10x^3 + \ldots\ldots \infty$$

multiply x

$$x S_\infty = 0 + x + 3x^2 + 6x^3 + \ldots\ldots \infty$$

$$S_\infty - x S_\infty = 1 + 2x + 3x^2 + 4x^3 + \ldots\ldots \infty \quad \text{A.G.P.}$$

$$S_\infty (1-x) = \frac{1}{(1-x)^2} \longrightarrow S_\infty = \frac{1}{(1-x)^3} \text{ Ans.}$$


xx
xx
xx
M = 1 + 2x + 3x^2 + 4x^3 + ... ∞
x
Mx = 0 + x + 2x^2 + 3x^3 + ... ∞
⊖
M - Mx = 1 + x + x^2 + x^3 + x^4 + ... ∞
G.P.
a = 1, r = x.
M(1 - x) = 1/(1-x) ⇒ M = 1/(1-x)^2

Acidic Question

Find sum of series :

$$S = 1 + \left(1+\frac{1}{5}\right)\left(\frac{1}{2}\right) + \left(1+\frac{1}{5}+\frac{1}{5^2}\right)\left(\frac{1}{2^2}\right) + \left(1+\frac{1}{5}+\frac{1}{5^2}+\frac{1}{5^3}\right)\left(\frac{1}{2^3}\right) + \ldots\ldots\infty$$

$\times\frac{1}{2}$ $\times\frac{1}{2}$ $\times\frac{1}{2}$

$\times\frac{1}{2}$

$$\frac{S}{2} = 0 + (1)\frac{1}{2} + \left(1+\frac{1}{5}\right)\left(\frac{1}{2^2}\right) + \left(1+\frac{1}{5}+\frac{1}{5^2}\right)\left(\frac{1}{2^3}\right) + \ldots\ldots\infty$$

(-)

$$\frac{S}{2} = 1 + \frac{1}{5}\times\frac{1}{2} + \frac{1}{5^2}\frac{1}{2^2} + \frac{1}{5^3}\frac{1}{2^3} + \ldots\ldots\infty$$

$a = 1,\ r = \frac{1}{5\times 2} = \frac{1}{10}$

$$\frac{S}{2} = \frac{1}{1-\frac{1}{10}} = \frac{10}{9} \Rightarrow \boxed{S = \frac{20}{9}}\ \text{fnf}$$

Question

If $|x| < 1$, then compute S_∞ :

$$S_\infty = 1^2 + 2^2x + 3^2x^2 + 4^2x^3 + \ldots\ldots \infty$$



Question [JEE Mains-2023]

A.G.P.

If $(20)^{19} + 2(21)(20)^{18} + 3(21)^2(20)^{17} + \cdots + 20(21)^{19} = k(20)^{19}$, then k is equal to

$$S = 1\,20^{19} + 2\,21.20^{18} + 3.21^2.20^{17} + \ldots\ldots\ldots + 19.(20)^1.21^{18} + 20.20^0 21^{19}$$

$\times \frac{21}{20}$ $\times \frac{21}{20}$

$$S = 1\,20^{19} + 2\,21.20^{18} + 3.21^2.20^{17} + \ldots\ldots\ldots + 19.(20)^1.21^{18} + 20.20^0\,21^{19}$$

$\times \frac{21}{20}$

$$\frac{21S}{20} = 0 + 1\,21.20^{18} + 2.21^2.20^{17} + \qquad\qquad + 19.21^{19} + 21^{20}$$

$$S - \frac{21}{20}S = \boxed{20^{19} + 21.20^{18} + 21^2.20^{17} + \ldots\ldots\ldots\ldots\ldots + 21^{19}} - 21^{20}$$

$$\frac{b^{11} - a^{11}}{b-a}$$

$$\frac{-S}{20} = \frac{21^{20} - 20^{20}}{21 - 20} - 21^{20}$$

$$\frac{-S}{20} = \frac{21^{20} - 20^{20}}{21 - 20} - 21^{20}$$

$$-\frac{S}{20} = \cancel{21^{20}} - 20^{20} - \cancel{21^{20}}$$

$$+S = +20^{21} = k(20)^{19}$$

$$k = 20^2 = 400 \text{ Ans.}$$

Question [JEE Mains-2014]

If $(10)^9 + 2(11)^1 (10)^8 + 3(11)^2 (10)^7 + \ldots. + 10(11)^9 = k(10)^9$, then k =





A. 100

B. 110

C. $\frac{121}{10}$

D. $\frac{441}{100}$

Question [JEE Mains-2023]

For $k \in \mathbb{N}$, if the sum of the series $1 + \frac{4}{k} + \frac{8}{k^2} + \frac{13}{k^3} + \frac{19}{k^4} + \cdots$ is 10, then the value of k is

• Multiple layer A.G.P.

• HW.

PROPERTIES OF SIGMA

Summation Σ

$$\# \; 1 + 2 + 3 + 4 + 5 + 6 = \sum_{r=1}^{6} r$$

Upper limit

$$\sum_{r=1}^{\square} r^2 =$$

$r = \square$ ← Lower limit.

Summation variable.

$$\sum_{r=-1}^{3} r^2 = (-1)^2 + 0^2 + 1^2 + 2^2 + 3^2$$

$$\sum_{r=1}^{5} (3r+1) = (3\times1+1) + (3\times2+1) + (3\times3+1) + (3\times4+1) + (3\times5+1)$$

$$= \sum_{r=1}^{5} 3r + \sum_{r=1}^{5} 1$$

$$(3\times1 + 3\times2 + 3\times3 + 3\times4 + 3\times5) + (1+1+1+1+1)$$

PROPERTIES OF SIGMA

① $\sum_{r=1}^{n} f(r) = f(1) + f(2) + f(3) \ldots\ldots + f(n)$

② $\sum_{r=1}^{n} (\text{const.})\, f(r) = (\text{const.}) \sum_{r=1}^{n} f(r)$.

③ Sigma add & sub. par distribute kr skte ho!

$$\sum_{r=1}^{n} \left(f(r) \pm g(r) \right) = \sum_{r=1}^{n} f(r) \pm \sum_{r=1}^{n} g(r).$$

Note:-

$$\sum_{r=1}^{3} \frac{2r+1}{r} = \frac{\sum_{r=1}^{3} (2r+1)}{\sum_{r=1}^{3} r}$$ ✗

Note :-

$$\sum_{n=1}^{3} n^3 = \sum_{n=1}^{3} (n^2 \cdot n) = \left(\sum_{n=1}^{3} n^2\right)\left(\sum_{n=1}^{3} n\right)$$ ✗

$1^3+2^3+3^3$

$(1^2+2^2+3^2) \cdot (1+2+3)$

PROPERTIES OF SIGMA

④ Sigma on any constant.

(i) $\sum_{r=1}^{3} 5 = 5+5+5 = 15.$ $\quad 5\times 3$

(ii) $\sum_{r=0}^{3} 5 = 5+5+5+5 = 20.$ $\quad 5\times 4$

(iii) $\sum_{r=-2}^{3} 5 = 5+5+5+5+5+5 = 6\times 5 = 30.$

$$\sum_{r=1}^{n} (\text{const.}) = n(\text{const.})$$

$$\sum_{r=0}^{n} (\text{const.}) = (n+1)(\text{const.})$$

PI-NOTATION

$\prod$ (used for multiplication)

$$\prod_{r=1}^{5} r = 1\times2\times3\times4\times5 = 120$$

$$\prod_{r=1}^{3} (2r-1) = 1\times3\times5 = 15$$

$$\prod_{r=0}^{3} 5 = 5\times5\times5\times5 = 5^4$$

IMPORTANT RESULTS

RAR

$$\sum_{r=1}^{n} r = 1 + 2 + 3 + \ldots . n = \frac{n(n+1)}{2}$$ = Sum of first n-Natural Numbers

$$\sum_{r=1}^{n} r^2 = 1^2 + 2^2 + 3^2 + \ldots . + n^2 = \frac{n(n+1)(2n+1)}{6}$$ = Sum of squares of first n-natural numbers

$$\sum_{r=1}^{n} r^3 = 1^3 + 2^3 + 3^3 + \ldots . + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$ = Sum of cubes first n-Natural Numbers

Proof :-

$$(r+1)^3 - r^3 = 3r^2 + 3r + 1$$

$r=1 \Rightarrow \quad 2^3 - 1^3 = 3\times 1^2 + 3\times 1 + 1$

$r=2 \Rightarrow \quad 3^3 - 2^3 = 3\times 2^2 + 3\times 2 + 1$

$r=3 \Rightarrow \quad 4^3 - 3^3 = 3\times 3^2 + 3\times 3 + 1$

$r=4 \Rightarrow \quad 5^3 - 4^3 = 3\times 4^2 + 3\times 4 + 1$

$\vdots$

$r=n \Rightarrow \quad (n+1)^3 - n^3 = 3\times n^2 + 3\times n + 1$

add

$$(n+1)^3 - 1 = 3(1^2+2^2+3^2+\ldots+n^2) + 3(1+2+3+\ldots+n) + n$$

$$(n+1)^3 - 1 = 3(1^2+2^2+3^2+\ldots+n^2) + 3(1+2+3+\ldots+n) + n$$

$$n^3+3n^2+3n = 3(\text{Req.}) + 3\,\frac{n(n+1)}{2} + n$$

$$n^3+3n^2+3n - \frac{3n(n+1)}{2} - n = 3(\text{Req.})$$

$$\frac{2n^3+6n^2+\cancel{6n}-3n^2-\cancel{3n}-\cancel{2n}}{2} = 3(\text{Req.}) = \frac{n(2n+1)(n+1)}{2}$$

$$\frac{2n^3+3n^2+n}{2} = \frac{n(2n^2+3n+1)}{2} = \frac{n(2n+1)(n+1)}{2}$$

Proof :-

$(n+1)^4 - n^4 =$ ~~~~~~~~~~

DO IT BY YOURSELF (DIBY)

GEOMETRIC PROGRESSION

DIBY-43

If the sum of an infinite GP a, ar, ar^2, ar^3, the squares of its each term is 150, then the sum of squares of ar^2, ar^4, ar^6, is: [JEE Mains-2021]

(A) $\frac{1}{2}$ (B) $\frac{25}{2}$ (C) $\frac{5}{2}$ (D) $\frac{9}{2}$

[Ans. : (A)]

DIBY-44

Three numbers are in an increasing geometric progression with common ratio r. If the middle number is doubled, then the new numbers are in an arithmetic progression with common difference d. If the fourth term of GP is $3r^2$, then $r^2 - d$ is equal to: [JEE Mains-2021]

(A) $7 + \sqrt{3}$ (B) $7 - \sqrt{3}$ (C) $7 + 3\sqrt{3}$ (D) $7 - 7\sqrt{3}$

[Ans. : (A)]

DIBY-45

If the sum of the second, third and fourth terms of a positive term. G.P. is 3 and the sum of its sixth, seventh and eighth terms is 243, then the sum of the first 50 terms of this G.P. is: [JEE Mains-2020]

(A) $\frac{2}{13}(3^{50}-1)$ (B) $\frac{1}{13}(3^{50}-1)$

(C) $\frac{1}{26}(3^{49}-1)$ (D) $\frac{1}{26}(3^{50}-1)$ [Ans. : (D)]

DIBY-46

The greatest positive integer k, for which 49^k+1 is a factor of the sum $49^{125}+49^{124}+\cdots+49^2+49+1$, is [JEE Mains-2020]

(A) 32 (B) 63 (C) 65 (D) 60

[Ans. : (B)]

DIBY-47

If the sum of the second, fourth and sixth terms of a G.P. of positive terms is 21 and the sum of its eighth, tenth and twelfth terms is 15309, then the sum of its first nine terms is: [JEE Mains-2025]

(A) 760 (B) 755 (C) 750 (D) 757

[Ans. : (D)]

DIBY-48

If three distinct numbers a, b, c are in G.P. and the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, then which one of the following statements is correct? [JEE Mains-2019]

(A) d, e, f are in A.P.

(B) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.

(C) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P.

(D) d, e, f are in G.P.

[Ans. : (C)]

DIBY-49

Find the sum of the following series

(i) 5 + 55 + 555 + ... to n terms

(ii) 0.3 + 0.33 + 0.333 + ... to n terms

[Ans. : (i) $\frac{5}{9}\left[10\left(\frac{10^n-1}{9}\right)n\right]$ (ii) $\frac{1}{3}\left[n-\frac{1}{9}\left(1-\frac{1}{10^n}\right)\right]$]

DIBY-50

If $x = 1 + a + a^2 + a^3 + \ldots\ldots \infty$ and $y = 1 + b + b^2 + b^3 + \ldots\ldots \infty$ show that

$1 + ab + a^2b^2 + a^3b^3 + \ldots\ldots \infty = \frac{xy}{x+y-1}$ where $0 < a < 1$ and $0 < b < 1$.

DIBY-51

For the two positive number a, b if a, b and $\frac{1}{18}$ are in a geometric progression, while $\frac{1}{a}$, 10 and $\frac{1}{b}$ are in an arithmetic progression, then $16a + 12b$ is equal to

[JEE Mains-2023]

[Ans. : 3]

DIBY-52

If the first 3 consecutive terms of a geometrical progression are the roots of the equation $2x^3 - 19x^2 + 57x - 54 = 0$ find the sum to infinite number of terms of G.P.

[Ans. : $\frac{27}{2}$]

DIBY-53

Let x_1, x_2, x_3, x_4 be in a geometric progression. If $2, 7, 9, 5$ are subtracted respectively from x_1, x_2, x_3, x_4 then the resulting numbers are in an arithmetic progression. Then the value of $\frac{1}{24}(x_1x_2x_3x_4)$ is: [JEE Mains-2025]

(A) 72 (B) 18 (C) 36 (D) 216

[Ans. : ....]

DIBY-54

If p arithmetic mean are inserted between 5 and 41 so that ratio $\frac{A_3}{A_{p-1}} = \frac{2}{5}$, then find the value of p.

[Ans. : 11]

DIBY-55

If $a_1 a_2 a_3 \ldots\ldots a_n$ are n arithmetic means inserted between 7 and 2015 whose sum is 56616, then

(A) n is 56

(B) n is 28

(C) $a_{19} = 2029/3$

(D) $a_{19} = 36543/57$

[Ans. : (C)]

DIBY-56

The sides of a right angled triangle are in arithmetic progression. If the triangle has area 24, then what is the length of its smallest side?

[Ans. : 6]

A.G.P.

DIBY-57

Let $S = 109 + \frac{108}{5} + \frac{107}{5^2} + \cdots + \frac{2}{5^{107}} + \frac{1}{5^{108}}$. Then the value of $\left(16S - (25)^{-54}\right)$ is equal to

[JEE Mains-2023]

[Ans. : 2175]

DIBY-58

Suppose $a_1, a_2, 2, a_3, a_4$ be in an arithmetico-geometric progression. If the common ratio of the corresponding geometric progression is 2 and the sum of all 5 terms of the arithmetico-geometric progression is $\frac{49}{2}$, then a_4 is equal to _____ .

[JEE Mains-2023]

[Ans. : 16]

DIBY-59

If $7 = 5 + \frac{1}{7}(5 + \alpha) + \frac{1}{7^2}(5 + 2\alpha) + \frac{1}{7^3}(5 + 3\alpha) + \cdots \infty$, then the value of α is :

[JEE Mains-2025]

(A) $\frac{6}{7}$ (B) $\frac{1}{7}$ (C) 6 (D) 1

[Ans. : (C)]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module:

It is never about end result

It's always about Journey

Thank You

#futureIITians